

AURA: A Hybrid CNN-Transformer Architecture for 3D Multimodal Brain Tumor Segmentation

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Abstract—This project presents AURA (*Attention · U-Net · Residual · All-modality*), a 3D hybrid CNN–Transformer model for automatic brain tumor segmentation. Trained on BraTS 2021 with a region-wise Dice + Focal loss, it reaches a mean Dice of 0.839 with strong calibration (ECE 0.100) and uncertainty estimation (AURC 0.175), clearly surpassing a plain residual U-Net baseline and matching a more complex eight-module variant overall while improving the clinically critical tumor core. At 45M parameters and 242 ms per scan, AURA is deployed as a browser-based clinical workstation that provides 3D rendering, per-region tumor volumes, confidence scores, and report export.

Index Terms—Brain Tumor Segmentation, Deep Learning, CNN-Transformer Hybrid, Multimodal MRI, Uncertainty Estimation

I. INTRODUCTION

Brain tumors are highly lethal, with a five-year survival rate of about 30%. MRI is the standard tool for diagnosis, where clinicians must delineate three tumor sub-regions—enhancing tumor (ET), tumor core (TC), and whole tumor (WT)—across four complementary modalities (T1, T1CE, T2, FLAIR), since no single modality shows the entire tumor. Done manually, this is slow: a radiologist contours the lesion slice by slice across all four modality volumes—on the order of 155 axial slices each—taking up to four hours per case, a process that is labor-intensive, subject to inter-rater variability, and offers little support for 3D visualization or instant volumetric measurement.

Automation is hard for two reasons. First, tumor voxels make up only about 1% of a scan, so a naive classifier collapses to predicting background. Second, tumors vary widely in size and location, leaving no positional shortcut. These properties demand careful loss design, balanced patch sampling, and a model that captures both fine local texture and global 3D context. Equally important in a clinical setting is *trustworthiness*: a model must not only be accurate but also signal when it is likely to be wrong.

This project addresses these challenges with **AURA**, a 3D hybrid CNN–Transformer model delivered as a complete end-to-end system rather than a model alone. Its goals are three-fold: **accuracy** across all three tumor regions, **trustworthiness**

through calibrated uncertainty that tracks real error, and **deployability** on a single consumer GPU. The expected benefit is a practical decision-support workstation that segments a scan in under a second and reports per-region volumes and confidence, turning hours of manual work into an interactive, browser-based clinical tool.

II. RELATED WORK

Automatic brain tumor segmentation has been driven largely by the annual BraTS challenge [1], [2]. Encoder–decoder CNNs remain the backbone of most leading entries: the U-Net [3] established the skip-connection design that fuses high-resolution detail with deep semantics, and nnU-Net [4] later showed that a carefully configured 3D U-Net—with the right preprocessing, loss, and training heuristics rather than architectural novelty—is still state of the art, topping several BraTS editions. Our Baseline follows this lineage as a plain 3D residual U-Net.

The Vision Transformer [5] motivated attention-based segmentation: UNETR [6] replaces the encoder entirely with a transformer, while hybrids such as TransBTS [7] and Swin-UNETR [8] insert windowed attention at selected resolutions. In 3D, however, the quadratic cost of self-attention scales with the cube of the spatial size, making fully transformer-based models data-hungry and heavy; AURA therefore keeps a convolutional backbone and adds self-attention *only* at the 8^3 bottleneck. Beyond accuracy, clinical use demands calibrated confidence: MC-Dropout [9] and test-time augmentation (TTA) yield predictive uncertainty without retraining, and the Expected Calibration Error (ECE) [10] and Area Under the Risk–Coverage curve (AURC) [11] quantify whether it is trustworthy—AURA integrates both and treats uncertainty as a first-class output rather than an afterthought.

III. METHODOLOGY

AURA is built as a complete end-to-end medical AI system, not just a model—raw MRI, preprocessing, model, training, evaluation, and a deployed web workstation are implemented,

trained, and evaluated together. The following subsections describe each component and the key implementation decisions behind it.

A. Dataset and Tumor Regions

We use the **BraTS 2021** dataset, which provides four co-registered MRI modalities per patient—T1, T1CE, T2, and FLAIR—each of shape $240 \times 240 \times 155$, with expert voxel-level labels for necrotic core (NCR), peritumoral edema (ED), and enhancing tumor (ET). Following the BraTS evaluation convention, these are grouped into three *nested* clinical regions: whole tumor (WT = NCR+ED+ET) $\supset$ tumor core (TC = NCR+ET) $\supset$ enhancing tumor (ET = {ET}). We split the 1,251 cases into 1,000 for training and 251 for validation (Table I).

TABLE I
DATASET SPECIFICATION

Property	Value
Dataset	BraTS 2021
Modalities	T1, T1CE, T2, FLAIR
Labels	0 BG · 1 NCR · 2 ED · 3 ET
Eval regions	ET, TC, WT (nested)
Volume shape	$240 \times 240 \times 155$
Train / Validation	1,000 / 251 cases

The four modalities are complementary: T1CE highlights the enhancing, contrast-taking tumor rim; FLAIR exposes the surrounding edema; and T1/T2 provide anatomical context. Because no single modality reveals the whole lesion, all four are acquired and fused at the input. The cohort is also highly heterogeneous in scale and position: whole-tumor volume spans roughly $3\text{--}362\text{ cm}^3$ (median $\approx 89\text{ cm}^3$, a $\sim 129\times$ range), and tumor centroids are scattered brain-wide with no consistent location. This rules out any positional shortcut a network might otherwise exploit and directly motivates the patch-based training and sliding-window inference described below.

Fig. 1 gives concrete examples of each labeled class across different patients, illustrating their distinct appearance and the difficulty the smaller, fragmented classes pose: necrotic core is patchy and embedded inside the core, edema is the largest and most diffuse, and enhancing tumor forms thin, irregular rims.

B. Preprocessing and Input Representation

Each modality is independently z-score normalized using only the mean and standard deviation of brain (non-zero) voxels, so background does not bias the statistics; the background is then re-zeroed. The four normalized modalities are stacked with a fifth **foreground mask** channel (1 where any modality is non-zero, 0 elsewhere), giving a **five-channel input** that lets the network distinguish true zero-signal tissue from skull-stripped background. The raw BraTS label 4 is remapped to 3, yielding a contiguous four-class target.

Because GPU memory cannot hold a full volume, the model trains on 128^3 **patches**. As tumor occupies only $\sim 1\%$ of

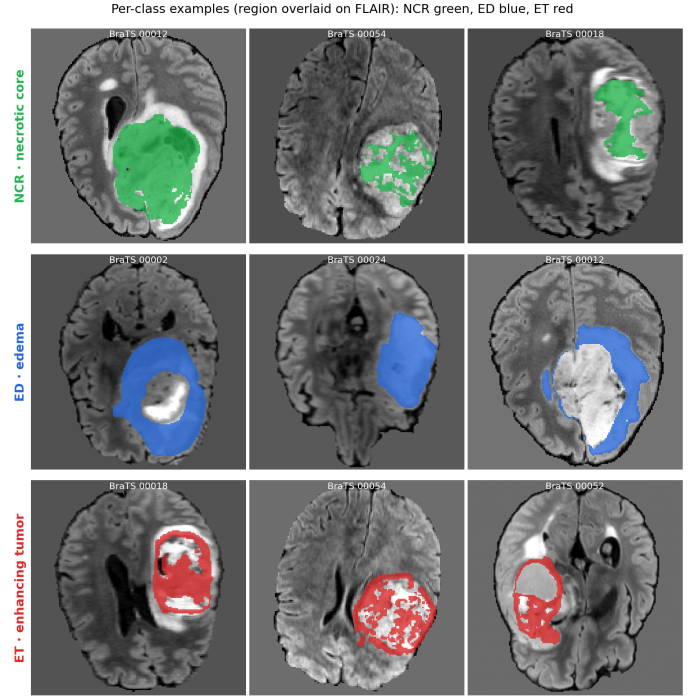


Fig. 1. Per-class examples (three patients per class), each region overlaid on its FLAIR slice: NCR (green), ED (blue), ET (red).

voxels, patches are sampled **50% tumor-centered and 50% uniformly random**: tumor-centered crops guarantee positive signal in every batch, while random crops preserve background context. Preprocessed volumes are cached as memory-mapped `.npy` arrays and tumor voxel coordinates are precomputed for fast sampling.

On-the-fly augmentations applied to each training patch include random zoom, per-axis flips, Gaussian noise and blur, and brightness/contrast jitter; the foreground-mask channel is excluded from intensity augmentations so it stays a clean binary signal. Volumes are stored as memory-mapped float16 arrays (cast to float32 at use) for fast, memory-bounded I/O.

C. The Class-Imbalance Problem

Segmentation here is severely imbalanced: only $\sim 1.07\%$ of voxels are tumor, so background versus tumor is roughly **99:1** and the enhancing tumor alone is about **412:1**. A plain cross-entropy loss collapses to predicting all-background yet still scores high voxel accuracy. This single property drives three core design choices—the region-wise Dice + Focal loss, tumor-centered sampling, and deep supervision.

D. Model Architecture

AURA is a 3D hybrid CNN–Transformer U-Net (Fig. 2). The intuition: convolutions excel at local texture while self-attention captures global 3D context, so we combine both but place the Transformer *only* at the deepest, lowest-resolution stage where it is most useful and computationally affordable.

- **Encoder (CNN)**. Four residual stages downsample $128^3 \rightarrow 8^3$ while widening channels $32 \rightarrow 64 \rightarrow 128 \rightarrow$

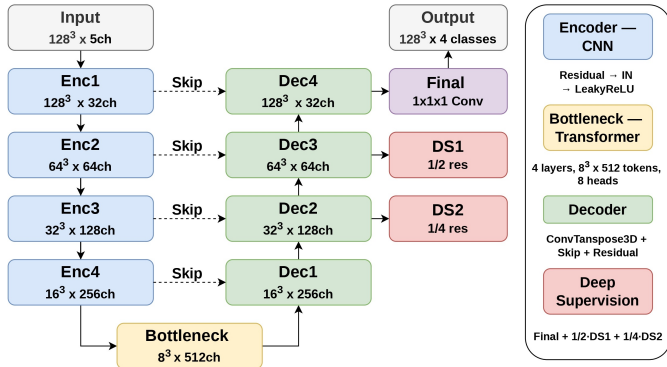


Fig. 2. AURA architecture: a residual-CNN encoder/decoder with a Transformer bottleneck at 8^3 and two deep-supervision heads (DS1 at 1/2, DS2 at 1/4 resolution).

256, with a 512-channel bottleneck. Each block uses Instance Normalization and LeakyReLU(0.01), stable at the small batch sizes 3D volumes force.

- **Bottleneck (Transformer).** At 8^3 resolution (512 tokens) a 4-layer, 8-head Transformer with relative position bias models global context. Restricting attention to 8^3 keeps its quadratic cost negligible and, crucially, keeps the model numerically stable under fp16—at higher resolutions the attention softmax tends to overflow in half precision.
- **Decoder.** Four mirrored stages use ConvTranspose3D upsampling, skip connections, and residual blocks to restore full 128^3 resolution.
- **Output head.** A $1 \times 1 \times 1$ convolution yields a **4-class softmax**. We chose softmax over an independent 3-channel sigmoid head because it enforces one label per voxel and avoids region-overlap contradictions (e.g. predicting ET but not WT).

The two deep-supervision heads (DS1, DS2) are used only during training, and decoder `Dropout3d` layers (rate 0.10) double as the stochasticity source for MC-Dropout at inference. The complete model has ≈ 45 M parameters (full profile in Table VI).

E. Loss Function

We optimize a **region-wise Dice + Focal loss** computed directly on the three clinical regions (WT, TC, ET) rather than the raw four classes, so the objective matches the evaluation metric. The Dice term [12] maximizes regional overlap and is robust to imbalance, while the Focal term [13] ($\gamma = 2$) down-weights easy voxels and concentrates gradient on hard, misclassified ones. **Deep supervision** adds the same loss at the 1/2- and 1/4-resolution decoder heads with decaying weights:

$$\mathcal{L} = \mathcal{L}_{\text{seg}}(\hat{y}) + \frac{1}{2}\mathcal{L}_{\text{seg}}(\hat{y}_{1/2}) + \frac{1}{4}\mathcal{L}_{\text{seg}}(\hat{y}_{1/4}) \quad (1)$$

$$\mathcal{L}_{\text{seg}} = \sum_{r \in \{\text{WT}, \text{TC}, \text{ET}\}} (\text{Dice}_r + \text{Focal}_r) \quad (2)$$

$$\text{Dice}_r = 1 - \frac{2 \sum p_r g_r}{\sum p_r + \sum g_r}, \quad \text{Focal}_r = -\sum (1 - p_t)^\gamma \log p_t \quad (3)$$

where p_r and g_r are the predicted probability and ground truth for region r , and the lower-resolution heads are supervised against nearest-neighbor downsampled targets.

F. Training Recipe

Training uses AdamW (lr $1e-4$) for 300 epochs. Because 3D patches limit the physical batch to 2, we use **gradient accumulation over 16 steps** for an effective batch of 32, and **fp16 mixed precision** with a gradient scaler to fit memory. An **EMA (decay 0.999)** of the weights yields the final model, and **gradient clipping (max-norm 1.0)** keeps training stable (Table II). The learning rate uses a **20-epoch linear warmup**—longer than a pure CNN needs, to ease in the bottleneck transformer under the small (~ 1 k-volume) regime—followed by cosine annealing to $1e-6$. A single AdamW group, a fixed seed (67), and best-model selection by mean validation Dice complete the recipe; the five highest-Dice EMA snapshots are retained for an optional snapshot ensemble. All development was done on a single consumer GPU; the same model that fits this hardware is the one deployed in the web workstation, so there is no train/serve gap.

TABLE II
TRAINING CONFIGURATION

Component	Setting
Optimizer	AdamW, lr $1e-4$
LR schedule	20-ep warmup $\rightarrow$ cosine ($\rightarrow 1e-6$)
Effective batch size	32 (batch 2 $\times$ accum 16)
Epochs	300
AMP	fp16
EMA decay	0.999
Gradient clipping	max-norm 1.0
Patch sampling	50% tumor- / 25% NCR-centered
Snapshots	top-5 EMA (by val Dice)

G. Inference and Evaluation Protocol

At inference, full volumes are processed with **sliding-window inference** [14] (128^3 window, Gaussian importance weighting, 50% overlap). We add **8-way flip test-time augmentation (TTA)** and a connected-component **post-processing** step that removes spurious small enhancing-tumor fragments. To quantify trustworthiness we estimate uncertainty via **MC-Dropout** (20 stochastic forward passes) and the spread across TTA views. Models are assessed along segmentation accuracy (Dice $\uparrow$, HD95 $\downarrow$ in mm), calibration (ECE $\downarrow$), and uncertainty (AURC $\downarrow$); pairwise comparisons use the paired Wilcoxon signed-rank test with bootstrap 95% confidence intervals.

Post-processing removes enhancing-tumor connected components smaller than 1,000 voxels and demotes them to necrotic core, suppressing false-positive specks that disproportionately hurt the small ET region. Calibration is reported under two masks: a brain-restricted ECE over all brain voxels,

and a stricter positive-only ECE over tumor voxels only—the latter far more honest under 99:1 imbalance, where a background-dominated ECE looks deceptively low.

H. Metric Definitions

All overlap metrics are computed per region (ET, TC, WT) as one-vs-rest binary problems at the voxel level, from the per-region true/false positive/negative counts (TP, FP, FN, TN):

$$\text{Prec} = \frac{TP}{TP+FP}, \quad \text{Rec} = \frac{TP}{TP+FN}, \quad \text{Spec} = \frac{TN}{TN+FP} \quad (4)$$

$$\text{Acc} = \frac{TP+TN}{TP+TN+FP+FN}, \quad \text{F1} = \frac{2TP}{2TP+FP+FN} \equiv \text{Dice} \quad (5)$$

Recall is equivalently the sensitivity, and F1 equals the Dice coefficient for binary masks—so our Dice columns *are* the per-region F1 scores. Because background dominates ($\sim 99\%$), accuracy and specificity are near-saturated; precision, recall, and Dice/F1 are the discriminating metrics.

I. Comparison Models

To isolate the architecture’s contribution, AURA is benchmarked against two references trained under the *identical* recipe: a **Baseline** (plain 3D residual U-Net, lower bound) and a **Complex** variant that stacks eight extra modules (modality stems, cross-modal attention, spectral Swin, frequency filtering, uncertainty and boundary heads, multi-scale fusion) as an upper-bound stress test. AURA adds only a single Transformer bottleneck to the baseline, letting us test whether a simpler design with far fewer added modules can match or beat a heavily over-engineered one. This is a comparison of *module count*, not parameter count: AURA ($\approx 45\text{M}$) is in fact larger than the Complex variant ($\approx 37\text{M}$) and roughly double the Baseline ($\approx 23\text{M}$), so “fewer modules” here means a simpler, more maintainable design rather than a smaller model.

IV. RESULTS AND EVALUATION

A. Quantitative Comparison

Table III reports validation performance. **AURA achieves the highest mean Dice (0.839)**, outperforming both the Baseline (0.825) and the more elaborate Complex model (0.833), and simultaneously attains the **best calibration (ECE 0.100)** and **best uncertainty (AURC 0.175)**. Per region, whole tumor is essentially saturated for all three models (≈ 0.92 – 0.93) and the three are within 0.024 on enhancing tumor, reflecting how hard the smallest sub-region is. AURA’s largest margin is on tumor core (TC, the surgical resection target): **+0.023 over Complex**, the most clinically meaningful gain in the table (per-region CIs and significance in Sec. IV-B).

B. Statistical Significance

Because point estimates alone can mislead, every AURA-vs-reference comparison is tested with a **paired Wilcoxon signed-rank test** over the 251 cases, alongside **bootstrap 95% confidence intervals (CIs)** on per-region Dice. AURA’s per-region Dice (TTA+post) is ET 0.785 [0.745, 0.823], TC 0.804 [0.768, 0.837], WT 0.928 [0.916, 0.939]. Against the **Baseline**, AURA’s gains are significant in every region (ET

+0.021, $p = 2.6 \times 10^{-4}$; TC +0.016, $p = 3.2 \times 10^{-6}$; WT +0.005, $p = 3.2 \times 10^{-6}$; mean +0.014, $p = 7.6 \times 10^{-9}$). Against the heavier **Complex** model the win is concentrated where it matters: TC improves by +0.023 ($p = 2.0 \times 10^{-4}$) and mean Dice by +0.006 ($p = 0.021$), while ET (-0.003 , $p = 0.12$) and WT (-0.001 , $p = 0.088$) are statistical ties. In short, AURA matches an eight-module model everywhere and *significantly* beats it on the clinically critical tumor core.

C. Per-Region Detailed Metrics

Table IV reports the full metric set for AURA. Precision and recall expose the per-region error asymmetry: TC is precision-heavy (0.935 vs. recall 0.783)—AURA rarely over-segments the core but misses parts of small or necrotic cores—whereas ET is balanced. Specificity is ≥ 0.999 everywhere (background is trivially easy) and AUC is uniformly ≥ 0.99 , confirming excellent probability ranking even where the thresholded Dice is lower.

D. Confusion Matrix and ROC Analysis

To characterize *class*-level errors rather than region overlap, Fig. 3 (left) shows the voxel-level 4-class confusion matrix, row-normalized so each row is the prediction distribution for one true class (i.e. per-class recall on the diagonal). Background, edema, and enhancing tumor are recovered well (ED 0.89, ET 0.90), but **necrotic core is the weak point: 0.54 recall, with 37% of NCR voxels misclassified as edema**—the quantitative root cause of the necrotic-core errors seen in the failure analysis. The ROC curves (right) show near-perfect ranking in all regions (ET/WT AUC 0.998, TC 0.992). Both panels use a single forward pass (baseline mode) pooled over all voxels, so the pooled ET recall here (0.90) is not directly comparable to the per-case, TTA+post recall in Table IV (0.836)—they are different averagings of different inference modes.

E. Qualitative Results

Fig. 4 (top) shows representative cases. AURA correctly recovers all three nested regions on typical tumors; failures concentrate where the enhancing tumor is very small or where necrotic tissue is ambiguous against edema—exactly the cases the model also flags as high-uncertainty. On typical tumors the model is highly accurate—for example, mean Dice 0.92 and 0.99 on cases BraTS2021_01418 (187,569 tumor voxels) and BraTS2021_01593 (77,802 voxels).

F. Failure Analysis

AURA’s errors are dominated by two patterns, both involving the smallest and most ambiguous tumors (Fig. 4, bottom). In case BraTS2021_01628 (whole tumor only 2,808 voxels) the enhancing tumor is missed almost entirely (ET Dice ≈ 0 , TC Dice 0.25); in BraTS2021_01508 (35,851 voxels) necrotic tissue is misread as edema (TC Dice 0.02). Both reflect the fundamental difficulty of the smallest sub-region under extreme imbalance. Crucially, these are exactly the cases the uncertainty head flags as low-confidence, which is what allows

TABLE III

VALIDATION RESULTS OVER ALL 251 CASES, TTA + POST-PROCESS MODE (BEST IN **BOLD**; ↓ LOWER IS BETTER). DICE/HD95 ARE PER-CASE MEANS; ECE IS THE POSITIVE-ONLY (TUMOR-VOXEL) VALUE; PER-REGION DICE 95% CIs AND SIGNIFICANCE ARE IN SEC. IV-B. AURA LEADS ON MEAN DICE, TC, ECE, AND AURC.

Model	Mean Dice	Dice ET	Dice TC	Dice WT	HD95 ↓	ECE _{pos} ↓	AURC ↓
Baseline	0.825	0.765	0.788	0.923	6.75	0.106	0.179
Complex	0.833	0.789	0.781	0.929	5.95	0.110	0.191
AURA	0.839	0.785	0.804	0.928	5.96	0.100	0.175

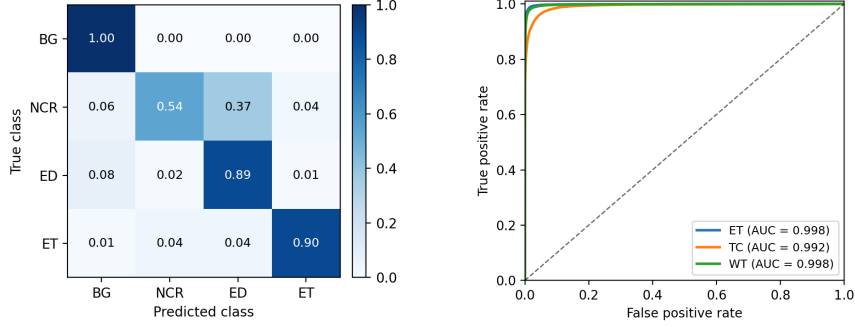


Fig. 3. (Left) Voxal-level 4-class confusion matrix (row-normalized); the dominant error is NCR→ED confusion (0.37). (Right) Per-region ROC curves (voxel-level, pooled); all regions reach $AUC \geq 0.99$.

TABLE IV

AURA PER-REGION DETAILED METRICS (TTA + POST-PROCESS). PRECISION/RECALL ARE PER-CASE MEANS OVER CASES WHERE THE REGION IS *present*; F1 IS THE PER-CASE DICE (= F1 PER CASE) AVERAGED OVER ALL 251 CASES. THE IDENTITY $F1 = 2PR/(P+R)$ HOLDS *per case* BUT NOT AFTER THESE DIFFERENT AVERAGINGS (MEAN OF A RATIO $\neq$ RATIO OF MEANS), SO THE COLUMNS ARE INTENTIONALLY NOT MUTUALLY RE-DERIVABLE.

Region	Precision	Recall	F1 (=Dice)	Spec.	AUC
ET	0.850	0.836	0.785	0.9998	0.995
TC	0.935	0.783	0.804	0.9998	0.992
WT	0.946	0.924	0.928	0.9994	0.999

selective prediction (next subsection) to recover accuracy by deferring them.

G. Calibration

Fig. 5 shows per-region reliability diagrams for the deployed TTA model (brain-restricted view). The accuracy bars stay close to the ideal diagonal—the brain-restricted ECE is only ≈ 0.007 —with a mild over-confidence gap in the highest-confidence bins, most visible for the tumor core. The stricter *positive-only* ECE in Table III (0.100), measured over tumor voxels alone, is the conservative headline figure; together the two views show a model that is well-calibrated overall while slightly over-confident on the hardest tumor voxels.

H. Trustworthiness

AURA assigns lower confidence to cases it is likely to get wrong, so its uncertainty is *actionable*. Under selective prediction (Table V), mean Dice rises monotonically as the least-confident cases are deferred, peaking at +0.019

(0.830→0.849) when the bottom 20% are skipped, before the shrinking sample adds noise at 70% coverage—exactly the behavior a clinician wants, with low-confidence cases routed for manual review rather than silently accepted.

TABLE V

SELECTIVE PREDICTION (TTA-ONLY MODE, NO POST-PROCESS/ENSEMBLE; HENCE THE 100%-COVERAGE 0.830 VS. TABLE III'S 0.839 TTA+POST). DEFERRING LOW-CONFIDENCE CASES RAISES DICE.

Coverage	Mean Dice
100% (all)	0.830
90%	0.846
80%	0.849
70%	0.842

I. Complexity and Deployment

At 45M parameters, 612 GFLOPs, and 242 ms per scan (4.1 cases/s, 1.9 GB GPU memory), AURA runs on a single consumer GPU (Table VI). The validated model is deployed as a browser-based clinical workstation (FastAPI backend, Niivue 3D viewer) that runs the same inference path used in evaluation and returns per-region volumes, confidence, anatomical localization, and an exportable report.

On a representative case it reports mean per-region confidence of WT 0.98, TC 0.99, ET 0.96. Because inference reuses the validated pipeline, the UI numbers match the reported results exactly—there is no train/serve gap.

V. CONCLUSION AND FUTURE WORK

We presented **AURA**, a 3D hybrid CNN–Transformer model for multimodal brain tumor segmentation, delivered as a

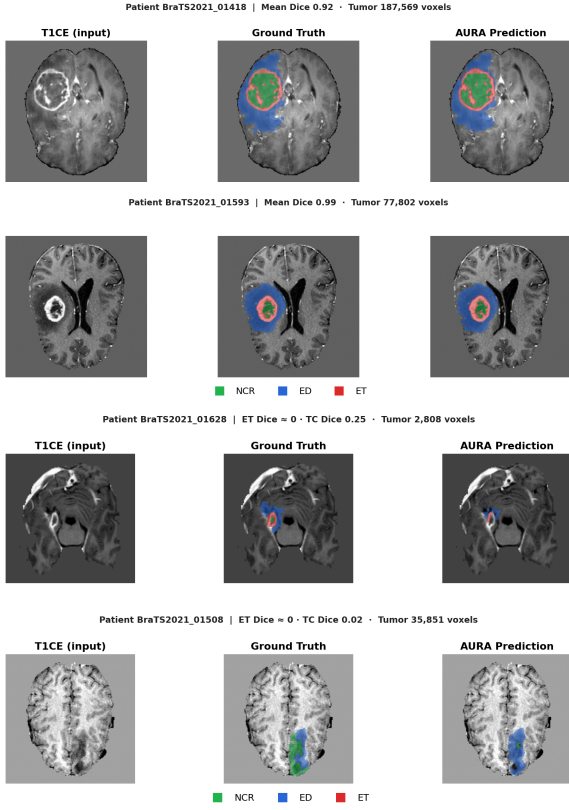


Fig. 4. Qualitative results (T1CE, ground truth, AURA prediction; NCR green, ED blue, ET red). *Top*: success cases—all three regions recovered. *Bottom*: failure cases—small or ambiguous tumors where ET is missed or NCR is read as edema, the same cases AURA flags as high-uncertainty.

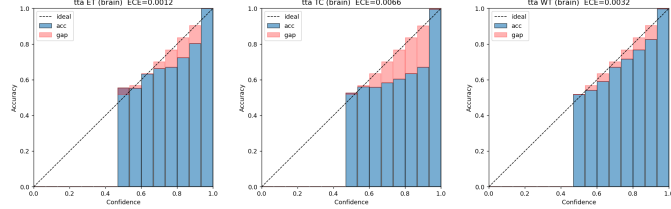


Fig. 5. Reliability diagrams (TTA) for ET, TC, WT: predicted confidence vs. empirical accuracy. Bars near the diagonal indicate good calibration; pink marks the over-confidence gap.

complete end-to-end system. AURA is (i) **accurate**—strong, balanced Dice across ET, TC, and WT; (ii) **trustworthy**—calibrated uncertainty that tracks real error and supports selective prediction; and (iii) **deployed**—the validated model runs live in a browser as a clinical workstation. Our results show that a single, well-placed Transformer bottleneck matches—and on tumor core beats—a model with eight stacked specialized modules.

Limitations. AURA is compared only against models we trained under one identical protocol, not against published standard baselines (e.g. nnU-Net or Swin-UNETR) in their authors’ configurations; and all results come from a *single* fixed train/validation split without cross-validation. The reported

TABLE VI
AURA COMPUTATIONAL COMPLEXITY

Metric	AURA
Parameters	45 M
FLOPs	612 GFLOPs
GPU memory	1.9 GB
Model size	172 MB
Inference time	242 ms
Throughput	4.1 cases/s

margins should therefore be read as indicative rather than definitive, and absolute numbers may shift under a different split or an externally tuned baseline.

Future work has two main directions: **external validation** across additional datasets and institutions to test generalization, and **model compression** via quantization and pruning for faster, cheaper inference on commodity hardware.

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REFERENCES

- [1] B. H. Menze *et al.*, “The multimodal brain tumor image segmentation benchmark (BRATS),” *IEEE Trans. Med. Imaging*, vol. 34, no. 10, pp. 1993–2024, 2015.
- [2] U. Baid *et al.*, “The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification,” *arXiv:2107.02314*, 2021.
- [3] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Proc. MICCAI*, 2015, pp. 234–241.
- [4] F. Isensee *et al.*, “nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation,” *Nature Methods*, vol. 18, pp. 203–211, 2021.
- [5] A. Dosovitskiy *et al.*, “An image is worth 16×16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021.
- [6] A. Hatamizadeh *et al.*, “UNETR: Transformers for 3D medical image segmentation,” in *Proc. WACV*, 2022, pp. 574–584.
- [7] W. Wang, C. Chen, M. Ding, H. Yu, S. Zha, and J. Li, “TransBTS: Multimodal brain tumor segmentation using transformer,” in *Proc. MICCAI*, 2021, pp. 109–119.
- [8] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. R. Roth, and D. Xu, “Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images,” in *Proc. MICCAI BrainLes Workshop*, 2021, pp. 272–284.
- [9] Y. Gal and Z. Ghahramani, “Dropout as a Bayesian approximation: Representing model uncertainty in deep learning,” in *Proc. ICML*, 2016, pp. 1050–1059.
- [10] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. ICML*, 2017, pp. 1321–1330.
- [11] Y. Geifman and R. El-Yaniv, “Selective classification for deep neural networks,” in *Proc. NeurIPS*, 2017, pp. 4878–4887.
- [12] F. Milletari, N. Navab, and S.-A. Ahmadi, “V-Net: Fully convolutional neural networks for volumetric medical image segmentation,” in *Proc. 3DV*, 2016, pp. 565–571.
- [13] T.-Y. Lin *et al.*, “Focal loss for dense object detection,” in *Proc. ICCV*, 2017, pp. 2980–2988.
- [14] M. J. Cardoso *et al.*, “MONAI: An open-source framework for deep learning in healthcare,” *arXiv:2211.02701*, 2022.